The Impact of Socioeconomic Factors on Schools' Average ACT Scores

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1 Abstract

The correlations between five socioeconomic factors (four census tract variables from the area the school is in (unemployment rate, percent of adults with a college degree, percent of children living in a married couple family, and median income), and the proportion of students of that school who qualify for free or reduced-price lunch) and average ACT score were calculated using single input and multiple input linear regression models.

The variable with the highest correlation and highest predictive power in regards to average ACT score was the percentage of students who qualified for free or reduced-price lunch.

2 Introduction

Is school performance correlated with socioeconomic factors?

Our approximation for school performance will be the average ACT score (or converted SAT score) for a particular public school in the United States.

A number of studies have been done relating ACT and SAT scores, socioeconomic background, and college admissions (Mineo, 2023) and have found that the socioeconomic background of a student is highly influential on their test scores (Geiser, 2020).

This sort of independent research is particularly valuable when the ACT Education Corp. is paying researchers to produce results that explicitly hide correlations between ACT scores and socioeconomic status (ACT, 2016).

The goal of this analysis is to examine the correlations between average ACT score and five socioeconomic factors. The socioeconomic factors in our dataset are four census tract variables from the area the school is in (unemployment rate, percent of adults with a college degree, percent of children living in a married couple family, and median income), and one is a direct measure of the proportion of students of that school who qualify for free or reduced-price lunch.

Note: a student qualifies for free or reduced price lunch if their family's income is below 130% or 185% of the Federal poverty line, respectively (USDA, 2025).

Two datasets were used: one from EdGap which contained the five socioeconomic factors, the school's average ACT score, and the National Center for Education Statistics (NCES) school ID; and one from the NCES containing additional data about each school, such as school type, location, and much more.

The much larger NCES dataset was used to provide more information for the smaller subset of schools found in the EdGap dataset. The year, ID, state, zip, school type, level, and charter status variables from the NCES dataset were merged with the EdGap dataset into a single dataframe using the corresponding school ID found in both datasets.

We would like to know specifically: how much do these school-wide factors affect school-wide ACT performance, and how do those effects hold up across school types (charter vs. non-charter) and across the 20 states available in this dataset?

3 THEORETICAL BACKGROUND

Initial visualizations through a pair plot of the five socioeconomic numerical predictor values and average ACT score suggested linear relationships, so linear regression models were chosen.

The fit of these linear models were confirmed as well.

4 METHODOLOGY

This analysis uses pandas (pd), numpy (np), matplotlib.pyplot (plt), seaborn (sns), sklearn, statsmodels.formula.api (smf), and statsmodels.api (sm). One optional map figure was created using plotly.offline (po) and plotly.graph_objs (pg).

Both datasets were imported, making sure that the datatype for school ID columns was object.

Basic cleaning was performed by deleting completely empty rows, converting percentage strings to floats and renaming columns to snake case.

Only the year, ID, state, zip, school type, level, and charter status columns from the NCES dataset were merged with the EdGap dataset into a single dataframe using the corresponding school ID found in both datasets.

Values that were out of range (negative ACT scores or percentages) were converted to NaN, all school levels other than high school were excluded, and rows missing data for average ACT score were dropped.

The NaN values for the remaining rows were imputed using an iterative imputer based on a regression relating to the other numerical variables in that row.

A correlation matrix was plotted using all five socioeconomic numerical predictor variables and the average ACT score (fig. 1).

Boxplots were used to identify outliers in the data. While there were outliers present, they were still within a plausible range so were not excluded.

Single predictor ordinary least squares regression models were performed for the median income, percent married parent households, and percent of students qualified for reduced-price lunch (fig. 2). P-values, R-squared, RMSE, and mean absolute error were also calculated for each model. Residual plots were also produced to assess the fit of the linear model.

A multiple variable ordinary least squares regression model using all five socioeconomic numerical predictor variables to predict the average ACT score was performed.

Variables with large p-values (percent married and median income) were dropped for the reduced variable model, which was created in the same way. Residual plots were also produced to assess the fits of both models. Mean absolute error and R-squared were calculated and were nearly identical across the full and reduced models. An ANOVA was also performed between the two models and found no significant differences.

A scaled reduced model was produced by normalizing the reduced number of numerical predictor variables. Mean absolute error and R-squared were calculated and were identical across the reduced and normalized reduced models.

A boxplot of average ACT scores per state was produced.

Individual dataframes for each state were created in order to find the correlation coefficient between percent reduced-price lunch and average ACT score for each state and then find the absolute difference between that coefficient and the one calculated for the whole dataset.

5 COMPUTATIONAL RESULTS

Figure 1. shows correlations between all five socioeconomic numerical predictor variables and the average ACT score. There are notable correlations between the variables themselves, as well as a strong, negative correlation between the percentage of students who qualify for free or reduced-price lunch and average ACT score.

Figure 2. shows that strong negative relationship between the percentage of students who qualify for free or reduced-price lunch and average ACT score.

Tables 1. shows the fit of the linear regression model for all five socioeconomic variables as predictors for average ACT score and table 2. shows that the fit did not change when median income and percent of children living in a married parent family were not used in the model.

Differences between charter and non-charter schools or between states were not obviously significant, so were not pursued further.

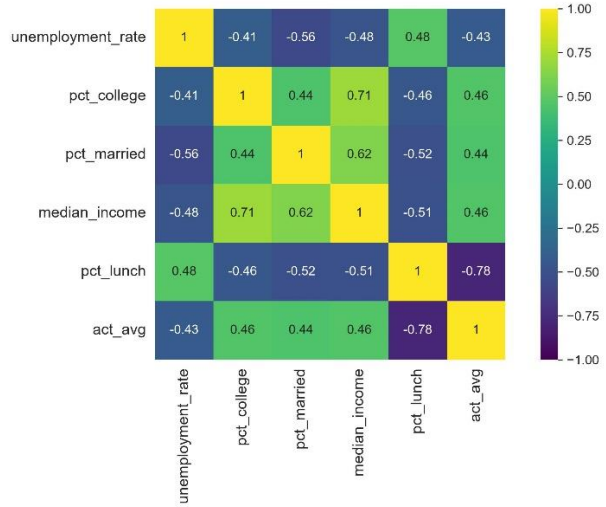


Figure 1: Correlation matrix between all numerical variables

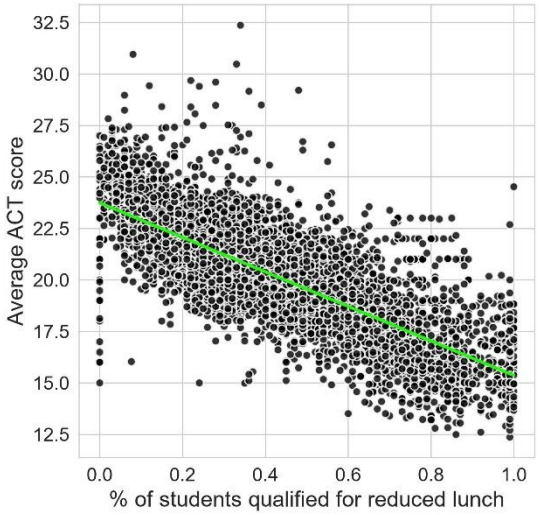


Figure 2: Scatterplot and linear regression line showing the relationship between the proportion of students who qualify for free or reduced-price lunch and average ACT score.

OLS Regression Results

Dep. Variable:	act_avg	R-squared:	0.627
Model:	OLS	Adj. R-squared:	0.627
Method:	Least Squares	F-statistic:	2432.
Date:	Sun, 18 Oct 2026	Prob (F-statistic):	0.00
Time:	19:48:38	Log-Likelihood:	-13333.
No. Observations:	7227	AIC:	2.668e+04
Df Residuals:	7221	BIC:	2.672e+04
Df Model:	5		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	22.6756	0.138	164.868	0.000	22.406	22.945
unemployment_rate	-2.2778	0.403	-5.650	0.000	-3.068	-1.487
pct_college	1.7450	0.158	11.070	0.000	1.436	2.054
pct_married	-0.0545	0.134	-0.408	0.684	-0.317	0.208
median_income	-1.658e-07	1.21e-06	-0.137	0.891	-2.54e-06	2.21e-06
pct_lunch	-7.5866	0.097	-78.400	0.000	-7.776	-7.397

Omnibus:	859.606	Durbin-Watson:	1.484
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3154.677
Skew:	0.570	Prob(JB):	0.00
Kurtosis:	6.030	Cond. No.	1.33e+06

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.33e+06. This might indicate that there are strong multicollinearity or other numerical problems.

Table 2: Shows the summary for the ordinary least squares model using all five socioeconomic variables as predictor variables.

OLS Regression Results

Dep. Variable:	act_avg	R-squared:	0.627
Model:	OLS	Adj. R-squared:	0.627
Method:	Least Squares	F-statistic:	4055.
Date:	Sun, 18 Oct 2026	Prob (F-statistic):	0.00
Time:	19:48:22	Log-Likelihood:	-13333.
No. Observations:	7227	AIC:	2.667e+04
Df Residuals:	7223	BIC:	2.670e+04
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	22.6324	0.103	220.747	0.000	22.431	22.833
unemployment_rate	-2.1990	0.373	-5.898	0.000	-2.930	-1.468
pct_college	1.7204	0.127	13.546	0.000	1.471	1.969
pct_lunch	-7.5720	0.092	-81.866	0.000	-7.753	-7.391

Omnibus:	861.313	Durbin-Watson:	1.484
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3153.034
Skew:	0.572	Prob(JB):	0.00
Kurtosis:	6.027	Cond. No.	25.8

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Table 1: Shows the summary for the ordinary least squares model using only the three socioeconomic variables that were shown to be significant by the first model as predictor variables.

6 DISCUSSION

All single input models showed the coefficients to be statistically significant, but only the model for reduced lunch (fig. 2) had a reasonably high R-squared value (0.613), meaning that it was the only single input model with a reasonably good fit.

A model using all five socioeconomic numerical predictor values had a high R-squared value (0.627) showing good fit, but also showed that two variables were not significant to the model (p-vales > 0.05) (table 1.). When those two variables were dropped to produce another model, the fit remained equally as high (R-squared = 0.627) (table 2.), confirming the insignificance of those variables in conjunction with the others.

That reduced variable model was then normalized to determine the strongest predictor of the remaining three. The normalized percent free or reduced lunch variable had a coefficient of -1.7746 compared to -0.1247 and 0.2837 for unemployment rate and percent of adults with a college degree respectively, showing that percent free or reduced lunch had, by far, the most influence on average ACT score.

This is likely due to the fact that the other four socioeconomic variables are census tract data and come from the areas surrounding the school and may, therefore, not be fully reflective of the student population because students may travel to other nearby districts or census tracts to attend school, whereas the proportion of students at a particular school who qualify for free or reduced-price lunch is a direct measure of the socioeconomic status of that school's student population.

7 Conclusions

These models demonstrate a strong negative correlation between the proportion of students at a school who qualify for free or reduced-price lunch and that school's average ACT score. We can also, therefore, conclude that as the proportion of students whose family's income is at or below 185% of the Federal poverty line rises, the average ACT score of that school will fall.

This demonstrates a direct relationship between poor economic background and poor ACT scores.

8 References

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